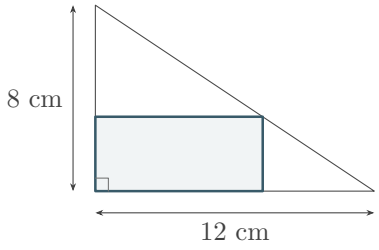
Optimization Worksheet

Part 1. Primer

Example 1. Among all pairs $x, y \geq 0$ satisfying $x+y=6$, which pair maximizes xy? Give the maximum value and justify why it is global.

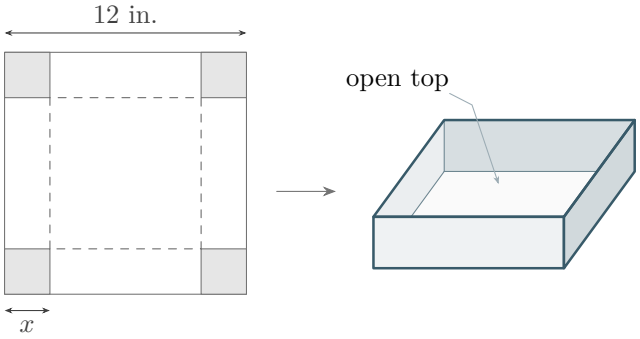
Example 2. A rectangle has area $36\,\text{cm}^2$. What is the smallest possible perimeter, and which dimensions attain it?

Practice 1. A rectangle sits inside a right triangle with legs of lengths 12 cm and 8 cm. Two sides of the rectangle lie along the legs, and its opposite corner lies on the hypotenuse. What is the greatest possible area of the rectangle?



Part 2. Optimization problems

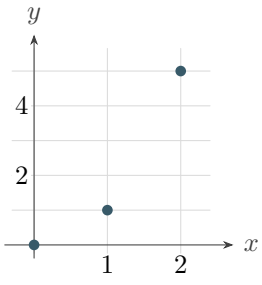
Example 3. Cut equal squares from the corners of a 12-inch square of cardboard. Fold up the four sides to make a box without a lid. What size cut gives the greatest volume? Give the box's dimensions and volume.



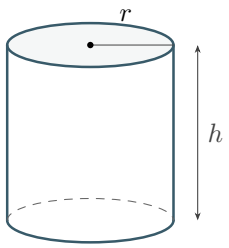
Example 4. A machine-learning model $y = mx+2$ is fitted to the points $(0,0)$, $(1,1)$, and $(2,5)$. With the intercept fixed at $b = 2$, its sum of squared errors is

$$E(m) = 4 + (m+1)^2 + (2m-3)^2.$$

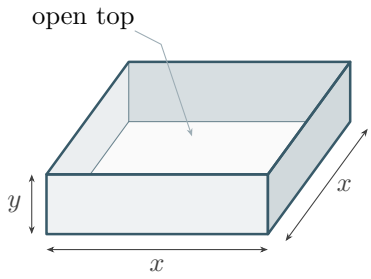
Which slope gives the least error, and what is that error?



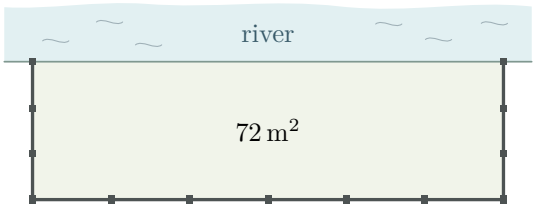
Practice 2. A closed cylindrical can must hold $54\pi\,\text{cm}^3$. Which radius and height minimize its total surface area, including both ends? What is that minimum area?



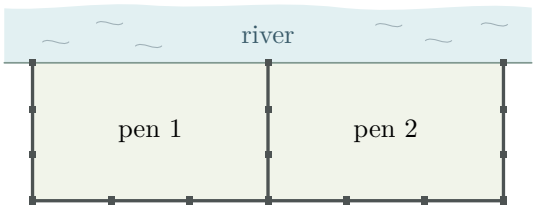
Practice 3. A tank has a square base of side x, height y, and no lid. Its base and four walls use $300\,\text{cm}^2$ of sheet metal. Which dimensions give the tank the greatest volume, and how much does it hold?



Example 5. A farmer needs a rectangular pasture of area $72\,\mathrm{m}^2$ beside a straight river. Only three sides need fencing. Which dimensions use the least fence, and how many meters are needed?

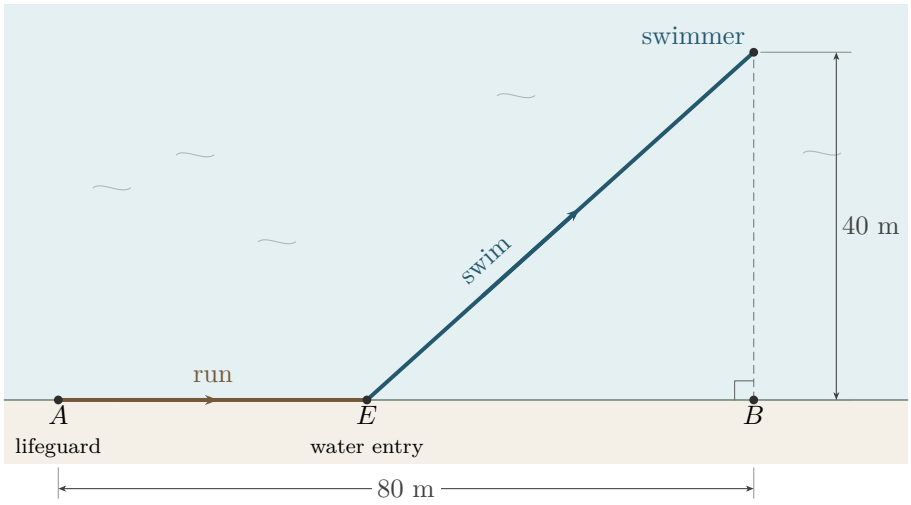


Practice 4. The farmer now wants two equal pens with the same total area of $72\,\mathrm{m}^2$. A divider runs perpendicular to the river. Which dimensions should each pen have to minimize the total fencing, including the divider? How much fence is needed?



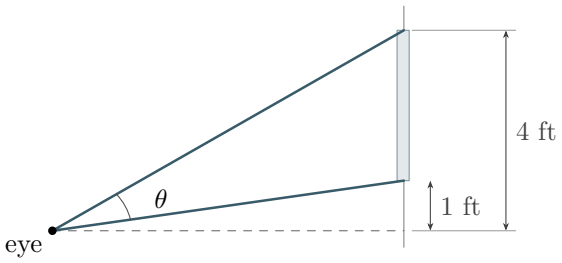
Part 3. Harder optimization problems

Example 6. A lifeguard at A spots a swimmer who may be drowning 40 m directly offshore from B. The straight stretch of beach from A to B is 80 m long. The lifeguard runs at 100 m/min and swims at 60 m/min in still water. Where should the lifeguard enter the water along $\overline{AB}$ to reach the swimmer fastest? What is the minimum time?



Practice 5. A band sells 120 tickets at \$10 each. It can raise the price in \$1 increments and expects to lose five sales per increment. The venue charges \$450 plus \$3 per ticket sold. Which ticket price maximizes revenue? Which price or prices maximize the band's profit?

Practice 6. A painting hangs on a vertical wall. Its lower edge is 1 ft above your eye level and its upper edge is 4 ft above. How far from the wall should you stand to maximize the angle between your sightlines to the two edges?



Practice 7. A 24-cm wire is cut into two pieces of positive length. One piece forms a circle and the other a square. Which cut makes their combined area as small as possible? Is there a cut that makes their combined area as large as possible?

